

LIGO NF-PE Calibration Pipeline

Project milestone: calibration drift in gravitational-wave parameter estimation across observing runs

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1. Motivation and progress

Question. Does a neural posterior estimator trained on simulated binary-black-hole (BBH) signals in LIGO O3a noise remain calibrated when deployed on O4a noise after a hardware/sensitivity upgrade? This matters because a fast model can retain plausible point estimates while becoming silently overconfident; incorrect credible intervals can propagate into astrophysical conclusions.

Progress. The team has implemented acquisition, quality vetting, waveform injection, whitening/bandpass/cropping, an SNPE model with a masked autoregressive flow (MAF), checkpointing, and Simulation-Based Calibration (SBC). The executed notebook contains preliminary O3a/O4a results; unit tests cover validation, injection alignment, PSD aggregation, checkpoint integrity, and calibration statistics.

Preliminary finding: O4a noise differs strongly from O3a, yet matched-size SBC comparisons show no strong evidence of additional cross-run calibration degradation. Distance is miscalibrated in both runs, so the result is not that the model is fully calibrated.

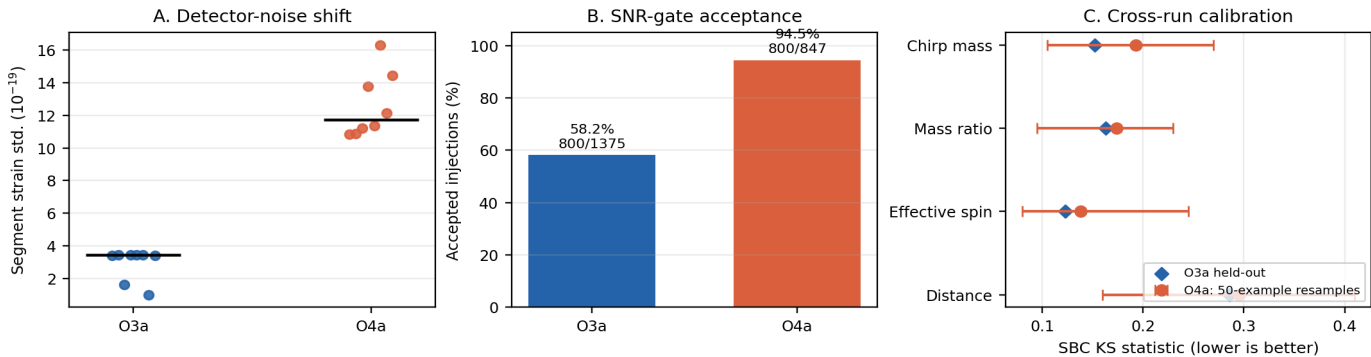
2. Data and experimental split

Stage	Representation and size	Allocation / role
Raw noise	H1 GWOSC strain; 8 segments/run x 1,024 s x 4,096 Hz = 33,554,432 samples/run (536.87 MB checkpoint/run)	O3a backgrounds for training; O4a backgrounds for distribution-shift testing
Accepted injections	800/run after SNR >= 8; O3a: 800/1,375 attempts; O4a: 800/847 attempts	Known BBH parameters injected into event-free real noise
Model-ready	Each example: 16,384 float samples (4 s at 4,096 Hz); target: 4 continuous parameters; processed checkpoint 209.77 MB	Implemented: 750 O3a train, 0 explicit validation, 50 O3a test, 800 O4a test
Planned revision	Same fixed test sets	Split the 750-development pool into 600 train / 150 validation; never tune on either test set

Inputs and targets. Input x is a whitened, 35-350 Hz bandpassed, merger-centered H1 strain window. Target $\theta = (\text{chirp mass, mass ratio, effective spin, luminosity distance})$. Source: public GWOSC H1 data; labels come from PyCBC IMRPhenomD simulations injected into that noise.

3. Exploratory data analysis

Exploratory and preliminary evidence from the executed pipeline



Panel	Interpretation
A - noise distribution	All eight O4a segment standard deviations exceed the O3a median. In 35-350 Hz, the ASD comparison gives KS=0.902 (p approximately 0) and a 65.1% median relative difference: a clear time/run-dependent covariate shift.
B - SNR selection	O3a rejects 41.8% of attempted injections versus 5.5% for O4a. The gate therefore changes the accepted target/SNR population across runs and must be audited as a potential selection effect.
C - calibration	For every target, the O3a held-out KS statistic lies inside the range from ten matched-size O4a resamples. This does not prove equivalence, but it provides no strong evidence of extra O4a degradation at n=50.

EDA coverage and next analyses

The executed notebook also inspects raw-strain time traces, strain histograms/statistics, merger-centered spectrograms, PSD/ASD curves, injection waveforms, SNR, and target ranges. Because the input is a 16,384-dimensional time series rather than a conventional feature table, raw Pearson feature-target heatmaps are not directly interpretable. The next EDA pass will compute correlations between physical targets and compact summaries (SNR, band powers, peak amplitude, duration), show target histograms/boxplots by run, and use permutation/ablation importance on the baseline and CNN. Derived chirp mass and mass ratio are expected to correlate because both are functions of component masses; this must be measured on accepted injections rather than assumed from priors.

4. Preprocessing and data challenges

Requirement	Implemented treatment / limitation
Missing values	No imputation: reject empty, wrong-length, NaN/Inf, gaps, or failed-data-quality segments; validation occurs at acquisition and checkpoint load.
Outliers	Exclude cataloged events (+/-1 h), require CBC data-quality bits, plausible strain std (1e-19 to 1e-17), and SNR >= 8. Training warned of localized >10x-IQR samples; compare robust clipping/no clipping only on validation data.
Scaling	Estimate run-specific Welch PSD; whiten strain; zero-phase Butterworth bandpass (35-350 Hz); crop 4 s around merger. SBI z-scores inputs/targets during training.
Categoricals	None are model inputs. Detector is fixed to H1; observing run defines train/test domain and is not encoded, preventing leakage.
Feature selection	Physics-guided time/frequency selection: merger-centered window and analysis band. The raw samples remain model input; engineered summaries are reserved for baseline/EDA.
Challenges	Large downloads/checkpoints, nonstationary noise, few real events, injection-selection shift, expensive rejection sampling, and one failed SBC example per run. Atomic checkpoints, bounded retries, explicit failure indices, and matched-size comparisons address these risks.

5. Methodology and planned experiments

Model	Why it is appropriate	Tuning and evaluation
Baseline: ridge / elastic-net on engineered summaries	Transparent, fast sanity check for whether band powers, SNR, amplitude, and duration contain predictive signal.	Tune regularization and L1 ratio on validation. Report standardized MAE/RMSE and R2 per target.
Improvement 1: 1D CNN point/heteroscedastic regressor	Learns local chirp morphology directly from strain; convolution adds translation-tolerant time-frequency feature extraction.	Tune channels, kernel sizes, dropout, learning rate, and weight decay. Compare point error plus interval coverage/NLL if heteroscedastic.
Improvement 2: SNPE + MAF posterior (implemented)	Returns the full conditional posterior needed for calibrated uncertainty, including non-Gaussian parameter degeneracies.	Tune flow depth/width, transforms, batch size, learning rate, and training simulations. Evaluate SBC KS, rank histograms, empirical 50/90% coverage, posterior width, and sampling-failure rate.

Experimental protocol. Use group-aware splits so windows sharing a noise segment never cross train/validation boundaries. Tune only on O3a validation data; freeze preprocessing and hyperparameters before evaluating the O3a and O4a test sets. Repeat training across seeds and report mean plus confidence intervals. Select the best model using validation performance with calibration as a first-class criterion, not test-set point error. Compare O3a versus matched-size O4a resamples to separate distribution shift from sample-size effects.

Preliminary results. SNPE+MAF converged after 39 epochs on 750 O3a examples. Valid SBC rows: 49/50 O3a and 799/800 O4a. O3a KS = 0.152/0.163/0.123/0.286 and matched-size O4a means = 0.193/0.174/0.138/0.295 for chirp mass, mass ratio, effective spin, and distance. Distance shows the clearest shared calibration problem. The PSD-drift monitor correctly flags the run shift, but the matched-size SBC evidence does not yet show additional O4a degradation.

6. Contributions and notebook ownership

Member	Specific contribution / notebook ownership status
Ali Lordifar	Repository integration, executable pipeline, documentation, proposal/PDF, tests, and current Git history. Declared owner: notebooks/main.ipynb.
Kennedy Goliman	Proposal feedback reflected in commit 4e4ab51 and learning-material review. Proposed sole owner: notebooks/learn_by_building.ipynb - team must confirm before submission.
Noah Fonck	Owner of notebooks/baseline_eda.ipynb: checkpoint audit, explicit O3a validation split, target/feature EDA, engineered summaries, NumPy ridge baseline, regression metrics, residual analysis, and permutation importance.
Aaron Rassi	No notebook ownership or committed contribution is evidenced in the current Git history. Assign a distinct numbered notebook and document completed analysis before submission.

Required pre-submission action: Git currently attributes existing commits to Ali; baseline_eda.ipynb is a new uncommitted contribution. The team must confirm Kennedy's proposed notebook ownership, add Aaron's distinct notebook contribution, execute notebooks so code-cell numbering starts at 1, and confirm the final ownership record. Repository: github.com/alilordifar/ligo-parameter-inference

Sources: repository README, config.py, TUTORIAL.md, PROPOSAL.md, notebooks/main.ipynb saved outputs, and Git history after pull on 2026-09-24; GWOSC data source: <https://gwosc.org/data/>.